Branching Fidelity: Testing World Models as Intuitive Physics Engines for Robot Safety

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Abstract: Before a robot executes an action, it should know whether its own uncertainty could change the outcome. People make this judgment with an intuitive physics engine [1]: they run a few noisy simulations over an uncertain scene and call it precarious when the outcomes disagree. We build this check for robots as a goal-free *fork probe*: sample the robot's measured perception and execution error, roll the intended action chunk forward 64 times, and flag it when the futures split. With a physics engine and a real camera-based perception model, the probe catches 88% of forks on unseen layouts with 7% false alarms. With a learned world model in place of the engine, a stochastic token-space model that predicts 3–4$\times$ better than a no-motion baseline catches only 18% of forks, and five further generative variants at most 3%: their sampled futures spread in every direction rather than along the branch. Deterministic DINO-WM-style dynamics that start each future from a perturbed scene, as the intuitive physics engine does, come relatively close (69% with 8% false alarms on held-out states; 68% with DINO-WM's predictor architecture) but plateau there. World models for safe deployment need this *branching fidelity*, which average-error benchmarks do not measure.

Keywords: World models, intuitive physics, uncertainty, safe deployment

1 Introduction

Manipulation policies, from diffusion policies to vision-language-action models, act by committing to short action chunks [2, 3, 4]. Deploying them safely requires a check before each chunk runs. Existing monitors address *epistemic* uncertainty, flagging inputs the model has not seen [5, 6]. We address a second kind that remains even for a perfect model: *physical* uncertainty. When a robot picks a block standing between two others, the same command can lift it cleanly or tip a neighbour depending on millimetres of execution error and perception error. Such a chunk is risky even if its nominal outcome is fine: whether it succeeds depends on errors the robot cannot control.

People handle this with an *intuitive physics engine* (IPE): a few approximate simulations over an uncertain scene, whose disagreement signals precariousness [1, 7, 8]. Our goal is to give robots the same ability, which needs two things: an honest model of the robot's own uncertainty, and a model to simulate with. Outside simulation the second must be learned, which gives the physical fidelity of world models a concrete, safety-relevant test: a world model is useful here only if its rollouts branch where the real dynamics branch, a property prediction error does not measure.

Contributions. (1) A *fork probe*: a pre-execution check that propagates the robot's measured perception and execution uncertainty through a rollout engine and flags splits, with no task labels, failure data or fitted thresholds (Sec. 2). (2) Validation with a physics engine and real camera perception on three unseen test sets, the reference a world model must reach (Sec. 3.1). (3) *Branching fidelity* as a world-model evaluation: generative world models that predict well on average fail it, while deterministic ones come close (Sec. 3.2).

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2 Setup and the fork probe

Task. Three Jenga-like blocks stand upright in a row; a simulated Franka Panda (MuJoCo) must pick the middle (red) block without toppling its neighbours. At each chunk start a policy commits to an 8-step action chunk; ours come from recorded episodes, but the probe treats the policy as a black box and sees only the chunk, one camera image and proprioception.

Fragility. A chunk is *fragile* if the robot's uncertainty, its execution error and its error in perceiving the scene, can send the outcome into qualitatively different regimes. This needs no goal or failure label, which a monitor around a generalist policy cannot assume. For our task we measure it physically: a state is a *fork* if between 2 and 62 of 64 futures sampled from this uncertainty topple a neighbouring block. A chunk that fails every time is not a fork and needs a task-level check.

Uncertainty model. Execution error comes from the robot's measured tracking residuals (median chunk maximum 3.2 mm). Perception is a frozen DINOv2 encoder [9] on one close-camera image with a small head that predicts the three blocks' poses (2.5 mm held-out error); its error model is measured once, offline, as a zero-mean Gaussian fitted to its errors on held-out images with known poses (a full 9×9 position covariance capturing correlated errors, plus rotation covariances).

Fork probe and rollout engine (Fig. 1a). The probe imagines 64 versions of the chunk. Each starts from the perceived scene shifted by one sampled perception error and runs the chunk with one sampled execution error; a 30-step imagined hold then lets the outcome settle. It flags the chunk if the 64 endings (object positions or image features) form two persistent groups: projected on their main axis of variation at hold steps 10 and 30, they fit two Gaussians better than one (BIC), the groups are well separated (Ashman's $D > 2$ [10]), the smaller has at least 2 members, and the grouping is the same at both times; these are fixed statistical conventions, not fitted to labels. A variant repeats this for 5 sampled scene estimates and flags if at least 3 split. The probe works with any rollout engine. Our reference is MuJoCo with the blocks at the sampled poses and their geometry, masses and friction from the simulator, which a real robot lacks; the learned alternative is a world model (Sec. 3.2).

Ground truth. One real outcome cannot grade a monitor, since a fragile chunk may succeed by luck. An oracle therefore samples the 64 futures from the *true* scene in exact physics and labels the state a fork as above. Two independent oracle runs give the same label on 99% of states. Each test set has unseen block layouts, about 60 forks and 100 safe states found by screening, and its protocol was frozen before the data existed.

3 Results

3.1 Propagating uncertainty through a physics engine works

First, the split test alone: from the true scene in exact physics, it flags all 23 forks among the 112 design states and 10% of the other 89. Second, real perception: a probe that trusts the pose estimate catches only 65–67% of forks, because perception errors (2–4 mm) are as large as execution errors; sampling the perception error raises this to 77–91% with 7–17% false alarms (Table 1). Third, images only: judging the split from DINOv2 features of the rendered endings instead of object positions changes little, catching 88% of forks (95% CI 77–94%) with 7% false alarms on test set C. This is the reference a world model must reach.

3.2 Learned world models: generative ones fail, deterministic ones come close

Without privileged geometry a learned model must take MuJoCo's role: given one image and the 64 perturbed chunks, do its rollouts branch where physics branches? Our model compresses DINOv2 tokens by a fixed per-patch PCA to a $28 \times 28 \times 8$ latent (applied to real futures, the split test on

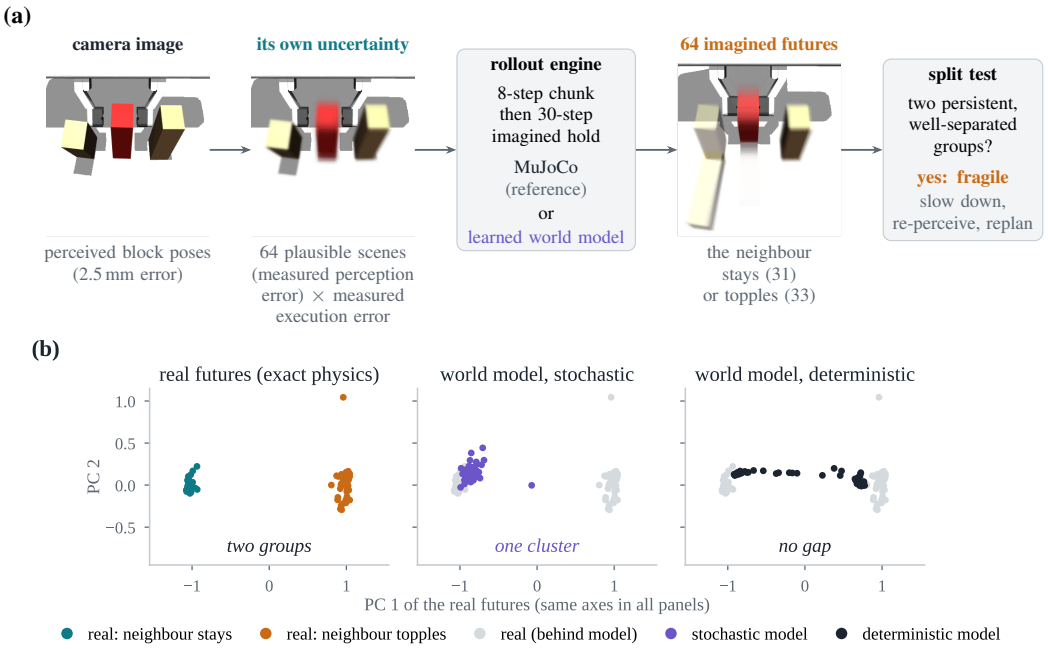


Figure 1: (a) The fork probe on one held-out state (exact renders; images average all 64 samples). (b) The same 64 futures in the world model's latent space, on the real futures' principal axes.

Table 1: Fork probe (MuJoCo, real perception) on unseen test sets: forks caught / false alarms (%).

outcomes read from object positions	Set A (153 states) caught	Set A (153 states) false	Set B (160) caught	Set B (160) false	Set C (156) caught	Set C (156) false
perception uncertainty ignored	65	23	65	12	67	24
fork probe	90	12	77	8	91	10
fork probe, 5 hypotheses	90	17	90	13	89	11
outcomes read from images: fork probe	88	12	77	8	88	9
5 hypotheses	–	–	–	–	**88**	**7**

this latent still catches 97% of forks). A convolutional network (six residual blocks conditioned on the action) predicts the next latent from the current one: the stochastic version is a flow-matching transition drawing fresh noise at every step, the deterministic control predicts it directly. Each training image comes with 8 real futures whose start scenes were perturbed by sampled perception error, so the spread the stochastic model learns is the robot's own uncertainty. Both beat a no-motion baseline on 19-step latent error by 3–4×, yet on test set C the stochastic model catches 18% of forks and the deterministic one 60%, against 88% for the engine (Table 2).

Where the uncertainty went wrong. Compared with real futures (Table 2), the stochastic model has the right *amount* of uncertainty but not its *shape* or *placement*: its futures spread equally in all directions and almost as much on safe states as on forks: it reproduces the average uncertainty, not where futures branch (Fig. 1b). The deterministic model's tiny spread is aligned with the branch axis and 32× larger on forks, suggesting, as in the IPE, that uncertainty should enter through the initial conditions. In follow-up experiments on 160 held-out states, generative models that produce whole clips, use an energy-score loss, see 4× more fork data or add an action-sensitivity loss catch 0–3% of forks, with the same all-direction spread. Putting the uncertainty into the initial conditions works instead: a learned operator perturbs the start latent by sampled perception error and the deterministic network rolls it out, catching 52% of forks at 7% false alarms, and 69% at 8% with 4× more fork data. This recipe plateaus at 68–72% with 3× longer training, more fork data, a 3-frame history,

Table 2: World models on test set C. Spread relative to real futures, its share on the main axis, and its ratio on forks vs other states (held-out states, 64 real futures each).

rollouts from	caught	false	spread	main axis	fork ratio
reference (MuJoCo; real futures)	88%	7%	1.00	0.57	4.8
token world model, stochastic	18%	2%	0.98	0.15	1.3
token world model, deterministic	60%	10%	0.04	0.69	32
5 generative variants (Sec. 3.2)	0–3%†	0–1%†	1.1–1.6	0.07–0.10	–
uncertain starts + deterministic, 4× forks	69%†	8%†	0.35	0.48	–

† on 160 held-out training-domain states (75 forks), not set C.

or DINO-WM's transformer predictor [11] instead of the convolutional network. With the same perturbed starts, a generative model still catches 0%, so the noise in its dynamics, not the missing starts, erases the forks.

4 Related work

Simulation under uncertainty. Beyond the IPE [12], robotics runs Monte Carlo rollouts over pose, friction or model parameters to score grasps and pushes [13, 14] or to build safety filters [15, 16]. Cannizzaro et al. [17] sample measured perception and placement noise through a simulator to predict whether a block tower stands, and Goyal [18] measure offline how fast perturbed chunks diverge. These predict a task outcome, a learned stability label [19] or a divergence rate. Closest in spirit, Zapolsky and Drumwright [20] sample model and sensor parameters to find divergent traces when validating plans offline, detecting divergence through task failure or inspection. Our probe runs before every chunk with measured errors and an automatic split test, a Monte Carlo test of proximity to a basin boundary [21].

Uncertainty-aware monitors. Latent safety filters (also on Jenga) and failure detectors use failure classifiers, epistemic or policy uncertainty [22, 5, 6, 23]: what a model does not know, not whether the physics is at a branch.

World models and their uncertainty. Learned simulators and latent world models [24, 25, 11, 26, 27] struggle with stiff contact [28]; audits find weak physical grounding [29, 30, 31]. Yu et al. [32] propose evaluating world models on branches created by different actions, by each branch's prediction error and action effect. Our branches instead arise from one action under small, unavoidable errors; there, our models have low prediction error but still miss the forks. Stochastic and ensemble models [33, 34] are evaluated by likelihood, calibration or return, none of which checks whether their spread splits where the dynamics branch.

5 Conclusion and limitations

A robot can judge fragility as people do, by propagating its own measured uncertainty, not just estimating it, through a rough simulator and checking whether the futures split. A physics engine with the scene's true parameters sets the bar, 88% of forks from one image. To be useful for safety monitoring and fine manipulation, learned world models should reach it, which takes branching fidelity: rollouts that split where the physics splits. Ours lack it despite low average error; deterministic dynamics recover much of it, reaching about 70% with uncertain starts and fork-rich data, but plateau there, so benchmarks should measure branching fidelity alongside average error. Closing the remaining gap is the next step: richer latents, proprioceptive inputs, fine-tuned pretrained world models such as V-JEPA 2-AC, and a real robot. *Limitations:* simulation, one task with three blocks; the engine probe uses the simulator's true shapes, masses and friction; one Gaussian models perception error for all images; test sets differ in difficulty; each world model was trained once; and the probe flags forks, not certain failures.

Use of generative AI. Generative AI tools (Claude) assisted with code, figures, literature search and drafting. The authors designed the study, checked all results and references, and take full responsibility for the content.

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